ITERATIVE LOCAL TANGENT ESTIMATION

Suppose a function generator sends an input waveform $f(t)$ to a piezo array. The piezo and the surrounding acoustic system transform this input into a sound wave, which is measured by a microphone. We write the measured signal as

$$y(t) = \mathcal{S}[f](t),$$

where $\mathcal{S}$ is the unknown nonlinear system. The symbol $\mathcal{S}$ represents everything between the function generator and the microphone: the piezo response, the acoustic propagation, and the microphone response.

Our goal is to choose $f(t)$ so that the measured signal $y(t)$ is as close as possible to a narrow pulse,

$$y_{\text{target}}(t) = A\delta_\epsilon(t - t_0),$$

where δ_ϵ is a narrow, experimentally realizable approximation to a delta function.

Discretizing the problem

In an experiment, both the input and output are sampled at discrete times. We therefore write

$$\mathbf{f} = \begin{pmatrix} f_1 \\ f_2 \\ \vdots \\ f_N \end{pmatrix}, \qquad \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{pmatrix}.$$

The nonlinear experimental system is now written as

$$\mathbf{y} = \mathcal{S}(\mathbf{f}).$$

Here $\mathbf{f}$ is the waveform sent to the function generator, and $\mathbf{y}$ is the waveform recorded by the microphone.

The desired microphone signal is

$$\mathbf{y}_{\text{target}} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ A \\ \vdots \\ 0 \end{pmatrix},$$

or, more realistically, a narrow Gaussian-like pulse centered at the desired arrival time.

We define the error vector

$$\mathbf{e} = \mathbf{y} - \mathbf{y}_{\text{target}}.$$

A natural objective function is

$$J(\mathbf{f}) = \frac{1}{2} \left\| \mathcal{S}(\mathbf{f}) - \mathbf{y}_{\text{target}} \right\|^2.$$

Equivalently,

$$J(\mathbf{f}) = \frac{1}{2} \sum_{i=1}^{N} (y_i - y_{\text{target},i})^2.$$

The goal is to find

$$\mathbf{f}^* = \arg\min_{\mathbf{f}} J(\mathbf{f}).$$

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The difficulty

The map $\mathcal{S}$ is nonlinear and unknown. Therefore we cannot simply invert it. Instead, we use the following idea:

> A nonlinear function is approximately linear if we look close enough.

At a current guess $\mathbf{f}^{(n)}$, we approximate the system by its local linear behavior:

$$\mathcal{S}\left(\mathbf{f}^{(n)} + \Delta\mathbf{f}\right) \approx \mathcal{S}\left(\mathbf{f}^{(n)}\right) + K_n \Delta\mathbf{f}.$$

Here K_n is the local tangent matrix, also called the Jacobian matrix. Its entries are

$$(K_n)_{ij} = \left.\frac{\partial y_i}{\partial f_j}\right|_{\mathbf{f}=\mathbf{f}^{(n)}}.$$

This number answers the question:

> If I slightly change input sample f_j, how much does output sample y_i change?

Estimating the tangent matrix experimentally

We do not know K_n analytically, so we measure it. First, send the current waveform $\mathbf{f}^{(n)}$ to the piezo and record

$$\mathbf{y}^{(n)} = \mathcal{S}\left(\mathbf{f}^{(n)}\right).$$

Now perturb one input sample at a time. Let $\mathbf{e}_j$ be the vector that is zero everywhere except at entry j:

$$\mathbf{e}_j = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Send the slightly perturbed waveform

$$\mathbf{f}^{(n)} + \epsilon\mathbf{e}_j$$

and record the output

$$\mathbf{y}^{(n,j)} = \mathcal{S}\left(\mathbf{f}^{(n)} + \epsilon\mathbf{e}_j\right).$$

Then

$$(K_n)_{ij} \approx \frac{y_i^{(n,j)} - y_i^{(n)}}{\epsilon}.$$

Doing this for all j gives the full local tangent matrix K_n.

$e_n = \vec{f}^{(n)} - \delta_n$

where $\delta_n = \begin{cases} 0, \text{ otw} \\ A, \ i \text{ s.t. } \vec{f}^{(n)}(i) = \max_j \{\vec{f}^{(n)}(j)\} \end{cases}$

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Finding the correction

Near the current waveform,

$$\mathbf{y}^{(n+1)} \approx \mathbf{y}^{(n)} + K_n \Delta \mathbf{f}.$$

We want this to be close to $\mathbf{y}_{\text{target}}$, so we choose $\Delta \mathbf{f}$ to minimize

$$\frac{1}{2} \left\| \mathbf{y}^{(n)} + K_n \Delta \mathbf{f} - \mathbf{y}_{\text{target}} \right\|^2.$$

Since

$$\mathbf{e}^{(n)} = \mathbf{y}^{(n)} - \mathbf{y}_{\text{target}},$$

this becomes

$$\frac{1}{2} \left\| K_n \Delta \mathbf{f} + \mathbf{e}^{(n)} \right\|^2.$$

The least-squares solution satisfies

$$K_n^T \left(K_n \Delta \mathbf{f} + \mathbf{e}^{(n)} \right) = 0.$$

Therefore

$$K_n^T K_n \Delta \mathbf{f} = -K_n^T \mathbf{e}^{(n)}.$$

If $K_n^T K_n$ is invertible, then

$$\Delta \mathbf{f} = -\left(K_n^T K_n \right)^{-1} K_n^T \mathbf{e}^{(n)}.$$

In practice, we usually regularize this equation:

$$\left(K_n^T K_n + \lambda I \right) \Delta \mathbf{f} = -K_n^T \mathbf{e}^{(n)}.$$

Thus

$$\boxed{\Delta \mathbf{f} = -\left(K_n^T K_n + \lambda I \right)^{-1} K_n^T \left(\mathbf{y}^{(n)} - \mathbf{y}_{\text{target}} \right)}$$

The parameter $\lambda > 0$ prevents the correction from becoming too large or too sensitive to noise.

Updating the waveform

We update the input waveform by

$$\boxed{\mathbf{f}^{(n+1)} = \mathbf{f}^{(n)} + \alpha \Delta \mathbf{f}}$$

where $0 < \alpha \leq 1$ is a step-size parameter.

A smaller α is safer when the system is strongly nonlinear, because the local linear approximation is only valid for small changes.

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Complete algorithm

Choose an initial waveform $\mathbf{f}^{(0)}$.

For $n = 0, 1, 2, \cdots$:

1. Send $\mathbf{f}^{(n)}$ to the piezo.
2. Measure $\mathbf{y}^{(n)}$.
3. Compute $\mathbf{e}^{(n)} = \mathbf{y}^{(n)} - \mathbf{y}_{\text{target}}$.
4. Experimentally estimate K_n.
5. Solve $\left(K_n^T K_n + \lambda I\right) \Delta\mathbf{f} = -K_n^T \mathbf{e}^{(n)}$.
6. Update $\mathbf{f}^{(n+1)} = \mathbf{f}^{(n)} + \alpha \Delta\mathbf{f}$.
7. Stop when $\mathbf{y}^{(n)}$ is sufficiently pulse-like.

Interpretation

The method does not require knowing the nonlinear function g. Instead, it uses the experiment itself to learn the local direction in which the input waveform should be changed.

At each step:

nonlinear system $\longrightarrow$ local linear approximation $\longrightarrow$ least-squares correction.

The system may be strongly nonlinear globally, but near a particular waveform it can often be approximated by a linear tangent map. The method repeatedly moves to a new waveform and then re-measures the tangent map there.

Thus the algorithm is essentially Newton's method, but applied to an experimentally measured waveform-to-waveform map.

Python-style pseudocode

```
# Goal:
# Find an input waveform f such that the measured output y
# is close to a desired narrow pulse y_target.

# Inputs:
#   f              initial waveform
#   y_target       desired microphone waveform
#   epsilon        small perturbation size
#   alpha          update step size
#   lambda_reg     regularization strength
#   max_iter       maximum number of iterations
#   tol            stopping tolerance

for n in range(max_iter):

    # 1. Send current waveform to experiment
    y = run_experiment(f)

    # 2. Compute output error
    error = y - y_target

    # 3. Check whether output is good enough
    if norm(error) < tol:
```

Why? could be max / other metric

```
        break

    # 4. Estimate local tangent matrix K
    # K[i, j] tells how output sample i changes
    # when input sample j is perturbed.
    K = zeros((N, N))

    for j in range(N):

        # unit vector in direction j
        ej = zeros(N)
        ej[j] = 1.0

        # perturb input waveform
        f_perturbed = f + epsilon * ej

        # measure output for perturbed waveform
        y_perturbed = run_experiment(f_perturbed)

        # finite-difference estimate of column j of K
        K[:, j] = (y_perturbed - y) / epsilon

    # 5. Solve regularized least-squares problem:
    #
    # (K^T K + lambda I) delta_f = - K^T error
    #
    A = K.T @ K + lambda_reg * eye(N)
    b = -K.T @ error

    delta_f = solve(A, b)

    # 6. Update waveform
    f = f + alpha * delta_f

    # 7. Enforce experimental constraints
    f = clip(f, -f_max, f_max)
    f = bandlimit(f, max_frequency)

# Final optimized waveform
f_optimized = f
```

Choices for the Initial Waveform

The iterative local tangent method requires an initial guess for the input waveform,

$$\mathbf{f}^{(0)}.$$

Since the nonlinear system is initially unknown, the first waveform should be chosen to probe the system broadly while avoiding excessively large amplitudes that may drive the piezo into strong saturation.

The goal of the first waveform is not necessarily to create a sharp pulse. Rather, its purpose is to reveal how the system responds so that local tangent information can be estimated.

Several choices are possible.

1. Gaussian pulse

A simple initial choice is a Gaussian pulse:

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$$f^{(0)}(t) = A \exp\left[-\frac{(t-t_c)^2}{2\sigma^2}\right].$$

where

- A is the amplitude,
- t_c is the pulse center,
- σ controls the pulse width.

Advantages:

- simple and easy to generate,
- localized in time,
- reveals ringing behavior,
- useful for a first trial.

Disadvantages:

- limited frequency content,
- may not excite all resonances.

A Gaussian pulse is often useful when little information is known about the system.

2. Chirp signal

A chirp sweeps through a range of frequencies:

$$f^{(0)}(t) = A \cos\left[2\pi\left(f_0 t + \frac{k}{2}t^2\right)\right].$$

The instantaneous frequency is

$$f_{\text{inst}}(t) = f_0 + kt.$$

Thus the signal gradually explores different frequencies.
A windowed chirp is often preferable:

$$f^{(0)}(t) = A \exp\left[-\frac{(t-T/2)^2}{2\sigma^2}\right] \cos\left(2\pi f_0 t + \pi k t^2\right).$$

Advantages:

- probes a broad frequency range,
- reveals resonances,
- useful for systems with memory,
- commonly used in acoustics and radar.

Disadvantages:

- longer duration,
- can excite nonlinear effects if amplitude is too large.

For the present problem this is often an excellent first choice.

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3. Random binary sequence

Another possibility is a random sequence:

$$f_n^{(0)} = As_n$$

where

$$s_n \in \{-1, +1\}$$

are randomly chosen.
Advantages:

- excites many frequencies simultaneously,
- widely used in system identification,
- useful for estimating local tangent matrices.

Disadvantages:

- physically less intuitive,
- may contain high-frequency components that stress hardware.

Pseudorandom binary sequences are common in experimental system identification because of their broad spectral content.

4. Random-phase multisine

Another commonly used signal is a sum of sinusoidal components:

$$f^{(0)}(t) = \sum_{m=1}^{M} A_m \cos(2\pi f_m t + \phi_m)$$

where the phases are chosen randomly:

$$\phi_m \sim U(0, 2\pi).$$

Advantages:

- frequency content can be controlled,
- nonlinear harmonics can be separated,
- widely used in Volterra system identification.

Disadvantages:

- somewhat more complicated to generate,
- parameter choices require care.

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Practical recommendation

For a strongly nonlinear piezo system with unknown behavior, a useful strategy is:

broad frequency coverage + moderate amplitude

rather than using a large-amplitude pulse immediately.
One practical approach is:

1. Start with a low-amplitude chirp.
2. Estimate the first local tangent matrix.
3. Perform iterative optimization.
4. Gradually increase waveform complexity if necessary.

The purpose of the initial waveform is not to solve the problem immediately. Its role is to provide enough information for the experiment to determine the local behavior of the nonlinear system.
Thus one should think of the first waveform as an exploratory probe rather than as a final candidate solution.



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